Prediction Model for Restaurant Health Inspection Midterm Report

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Repository: https://github.com/ahnzalacs/data1030-project

Introduction

In our DATA 1030 project, we chose to examine New York City restaurant Health Inspection data. None of us have prior experience with this topic, but we all thought it would be something fun to explore. Our data was gathered from the New York City Department of Health and Mental Hygiene via API request. While previous projects have used similar datasets, our approach solves different challenges. Nadler [1] used logistic regression to examine how restaurant type and location were associated with receiving a Grade A and found that some restaurant types had lower odds of receiving the highest inspection grade. Our project focuses on predicting whether an initial inspection will pass or fail. Our data focuses on the last 3 year window, containing inspections occurring from January 3rd 2023 to October 20th 2026. Our model will predict the outcome for the inspection (pass/fail) to guide health inspectors towards high-risk restaurants. Additionally, we plan to extend our model and test its performance across different cities' restaurant health inspection data. We aim to create a model that can improve health inspection systems globally.

Data

Our data comes from the DOHMH New York City Restaurant Inspection Results dataset, published by the NYC Department of Health and Mental Hygiene through NYC Open Data (dataset id 43nn-pn8j) [2]. Each record is a single violation cited during an inspection, along with

the restaurant's name, location, cuisine, inspection score and grade. Inspectors assign penalty points for every violation, so a higher score means worse conditions. A score of 0 to 13 is considered an A, and a score of 14 or more triggers a re-inspection that decides between a B (14 to 27) and a C (28 or more). The API keeps a rolling window of roughly three years and only lists currently active restaurants, so the data is biased toward restaurants that survived. We are also considering Yelp reviews as a supplementary source, since customer complaints about hygiene may capture signals not reflected in inspection history. We are currently evaluating whether reviews can be reliably matched to inspection records.

We collected the data in two stages. In the first stage we used a collection script (src/collect.py) to pull the full dataset from the API on October 20th, 2026. Since the API caps each request at 50,000 records, the script downloaded the data in pages and retried any failed requests, giving us a raw file of 295,422 rows and 26 columns. The pull time and row count are saved in a metadata file, so we can always trace which version of the data we used. Once we have the raw data we use a processing script (src/build_dataset.py) to turn the violation records into one row per inspection. We drop restaurants that haven't been inspected yet and treat zero-valued boroughs and coordinates as missing, and combine each inspection's violations into a score, violation counts and closure status. We then build history features using only earlier inspections, so nothing leaks from the inspection we're trying to predict. Finally, we keep only initial cycle inspections from 2023 onward, which leaves 44,890 inspections of 22,045 restaurants. Figure 1 shows how many rows remain at each stage.

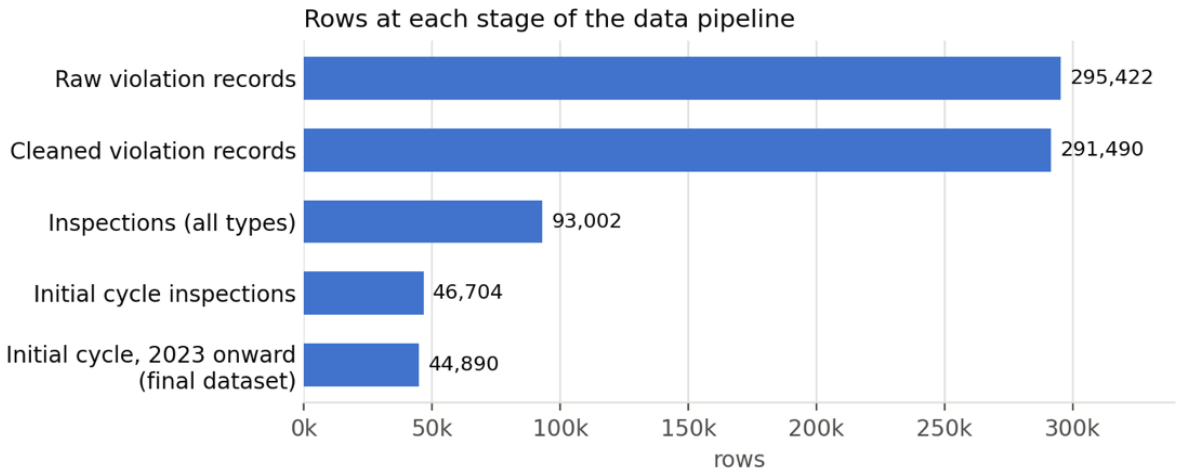


Figure 1. Number of rows at each stage of the data pipeline, from the raw violation records to the final modeling dataset.

The final dataset has 44,890 rows and 24 columns: 3 identifiers, 19 features and 2 candidate targets. We plan to focus on using the fail variable as our main target variable. Detailed documentation for the raw and processed datasets are available in (data/README.md) in our GitHub repository. Each row is one restaurant inspection from the timeframe specified above.

Group	**Columns**	**Type**
Identifiers (not features)	camis, dba, inspection_date	id / text / date
Restaurant facts	boro, zipcode, cuisine_description	categorical (5 boroughs, 89 cuisines)
	latitude, longitude	continuous (degrees)
	is_chain	binary

Timing	inspection_year, inspection_month, inspection_dayofweek	ordinal
History	n_prior_inspections, n_prior_initial_inspections, n_prior_critical_violations, n_prior_closures	count
	prev_initial_score, mean_prior_initial_score, max_prior_initial_score	continuous (points)
	days_since_prev_inspection, days_since_prev_initial	continuous (days)
	prev_initial_failed	binary
Targets	fail (main), score (alternative)	binary / continuous

Rows are not independent, since restaurants appear between one and five times (about two on average), so the train and test sets should be split by restaurant. The data is also time-ordered, with roughly 12,500 to 13,000 inspections per year from 2024 onward; 2023 has fewer because the rolling window retains less of that year (Figure 2).



Figure 2. Left: number of restaurants by how many times they appear in the dataset. Right: number of inspections per year.

Missing values are mostly structural. Features based on a previous initial inspection are missing for 46.3% of rows, and days since any previous inspection for 27.6%, because those restaurants have no earlier inspection in the data window. Location, zip code and borough are missing for under 2% of rows.

Our main target is a binary indicator of failing the initial inspection, defined as a score of 14 or more. This matches the real decision at the inspection: an immediate A or a required re-inspection. About 40.1% of inspections fail, so the classes are moderately imbalanced. We keep the raw score as an alternative target for a possible regression framing; it is right-skewed, with a median of 13 and a maximum of 157. The most important features describe a restaurant's track record: its previous initial score and whether it failed, the average and worst earlier scores, prior critical violations, prior closures and the time since its last inspection. The remaining features describe the restaurant (cuisine with 89 categories, borough, location and a chain flag for names at five or more locations, covering 20.4% of inspections) and the timing of the inspection (year, month and day of week).

Exploratory Data Analysis

We explored several features that may be related to whether a restaurant fails its current initial inspection. In our dataset, about 40.1% of the inspections resulted in failure, while 59.9% passed. This shows that the two classes are not extremely imbalanced.

We first looked at failure rates across different cuisine types. To avoid comparing cuisine categories with very few observations, we only included cuisines with at least 300 inspections. As shown in Figure 3, there are large differences in failure rates across cuisine types. Some of the highest-failure cuisines have failure rates above 50%, while some of the lowest are below 25%. This suggests that cuisine type may be related to inspection outcomes and could be useful as a feature in our model.

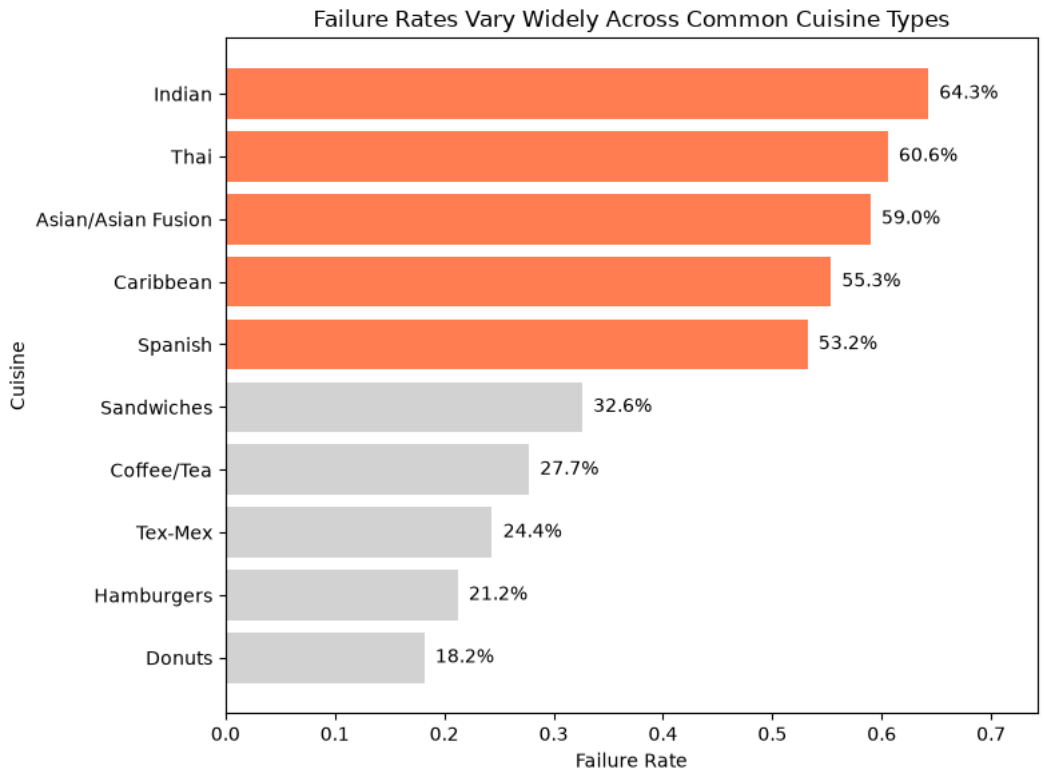


Figure 3. Inspection failure rates differ greatly across cuisine types.

We also examined whether a restaurant's previous inspection performance is related to its current inspection result. Figure 4 shows a clear pattern between the previous initial inspection score and the current failure rate. Restaurants with previous scores between 0 and 5 have a current failure rate of about 21.4%. The failure rate increases as the previous score increases, reaching 58.6% for restaurants with previous scores of 31 or higher. This suggests that a restaurant's past inspection performance may be an important predictor of its future inspection outcome.

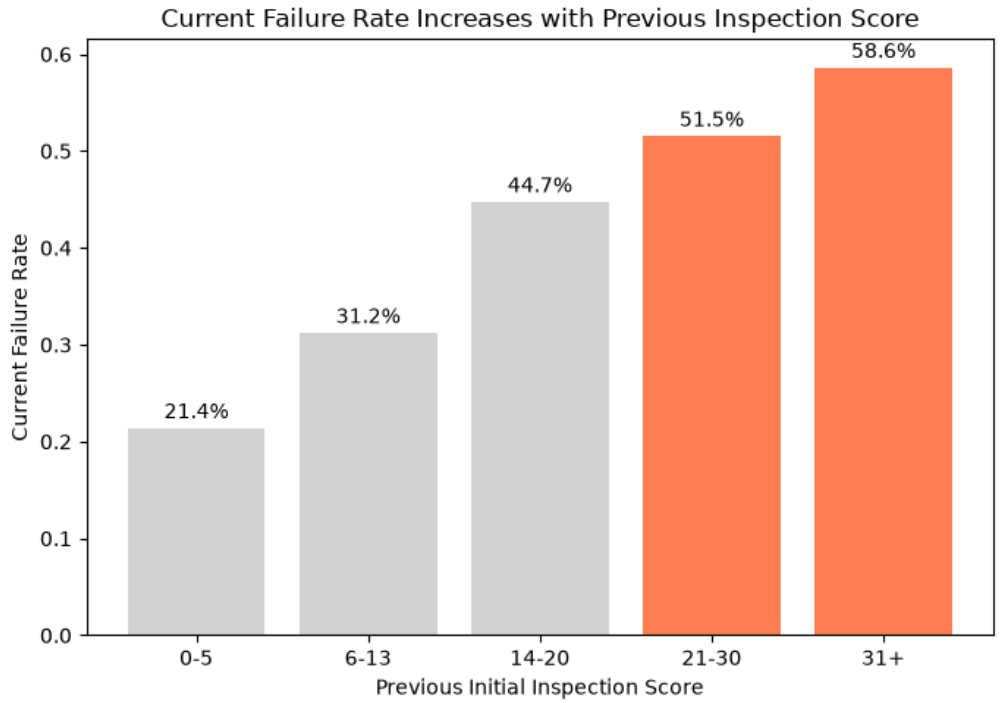


Figure 4. Inspection failure rates increase with previous initial inspection scores.

We also looked at whether inspection outcomes differed between chain and non-chain restaurants. Figure 5 shows a noticeable difference between the two groups: about 43.6% of non-chain restaurants failed their inspections, compared with 26.2% of chain restaurants. This

suggests that whether a restaurant belongs to a chain could be relevant when predicting inspection outcomes.

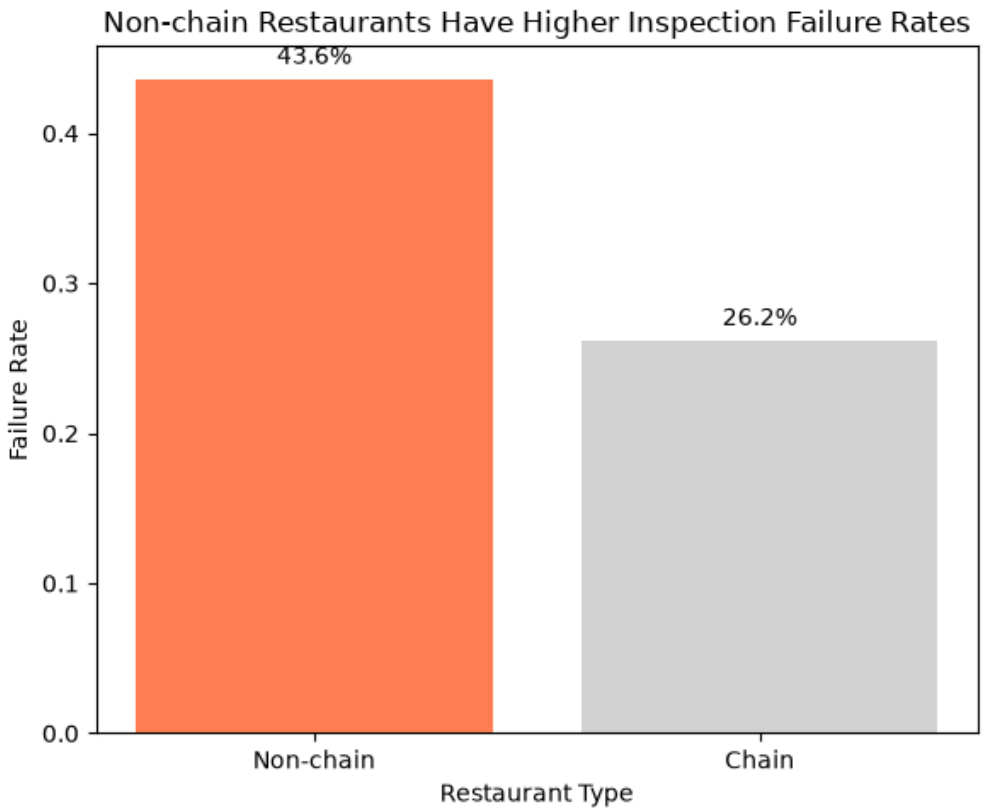


Figure 5. Restaurant chain status informs inspection failure rate

Overall, the EDA shows that restaurant characteristics, such as cuisine type, chain status and previous inspection performance, are associated with the current inspection result. These features will be considered as potential predictors when we develop our machine learning models.

Problem description and potential impact

By analyzing this past health inspection data, we hope to use a supervised machine learning model to extrapolate insights about the likelihood of passing the inspection. In our dataset, our target variable is represented by the 'fail' column. A value of 0 equates to a passing

inspection, while a value of 1 equates to a failed inspection. Thus, our model is a binary classification. When running our model, we will exclude restaurant names because they serve as identifiers. We also plan to exclude boroughs as a predictor because it is specific to New York City and may limit the model's ability to generalize restaurant inspection data from other cities. Predictions about the outcome of the inspection can inform the public servants about which restaurants should be inspected more frequently than others. By targeting restaurants with predicted failed inspections, Health Inspection teams can optimize their scheduling, maximize their resource allocation, and develop a system that proactively protects public health. This will ultimately lead to a smarter and safer inspection system for cities worldwide.

References

[1] Nadler, D. (2014). Using logistic regression to model New York City restaurant grades over a two-year period. Epidemiology, Biostatistics, and Public Health, 11(3). https://doi.org/10.2427/9442

[2] New York City Department of Health and Mental Hygiene. (n.d.). DOHMH New York City restaurant inspection results [Data set]. NYC Open Data. https://data.cityofnewyork.us/Health/DOHMH-New-York-City-Restaurant-Inspection-Results/43nn-pn8j